

ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA (REGD.)

Eighteenth All India Talent Competitions in Mathematics
January 1987

the Question Booklet until you are told to do so.

Multiple choice Test. Each question is followed by four answers (a), (b), (c), (d). Only one of them is correct. When you have decided on your choice, write the appropriate letter on your answer sheet.

Do not write more than one answer. Such responses will not secure marks.

If you are unable to solve a problem, it is better not to answer the question. Avoid guessing as you are penalized one-fourth the value of the question for a wrong answer.

Make sure that you understand thoroughly the coding system for your answer sheet. If you are not sure, ask your supervisor to clarify it.

Only rough-paper is permitted to use rough-paper. **No other aids** (like the instrument box, etc.) are permitted.

Diagrams are **not** drawn to scale. They are intended as aids only.

Before beginning to write your answers, fill in the details of your bio-data (Name, Age, etc.) in the appropriate boxes you find in the answer sheet.

At the end of the competition, return only the answer sheet. The question - booklet and the answer sheet may be retained by you.

When your supervisor instructs you to begin, you have 60 minutes of working time.

all divisors of 30, excluding 30 is

- a) 24
- b) 22
- c) 44
- d) 42

$3\sqrt{3} - 4\sqrt{4}$ lies between

- a) 1 and 2
- b) $\sqrt{2}$ and $\sqrt{3}$
- c) .02 and .03
- d) 3 and $2\sqrt{3}$

$\sqrt{2}$, value of x is

- a) $\frac{1}{4}$
- b) $\frac{1}{6}$
- c) $\frac{1}{\sqrt{2}}$
- d) $\frac{1}{3}$

root of $4\frac{25}{36}$ is

- a) $2\frac{5}{6}$
- b) $\frac{10}{6}$
- c) $\frac{13}{6}$
- d) none of these

number subtracted from $\frac{1}{4}$ of its value is
by $\frac{1}{3}$ gives the answer $\frac{1}{2}$. What answer
if you divide the number by $\frac{1}{6}$?

- a) 72
- b) 30
- c) 180
- d) $\frac{1}{2 \times 3 \times 4 \times 5 \times 6}$

7 terms of $1-1+1-1+\dots$

- a) 1
- b) -1
- c) 147
- d) -147

the two unequal sides of an isosceles triangle is

- a) 2
- b) $\frac{1}{\sqrt{2}}$
- c) $2\sqrt{2}$
- d) 1

10 is divisible by 11 then * is

- a) 0
- b) 1
- c) 5
- d) 9

of first n natural numbers is

- a) n
- b) $\frac{n}{2}$

nt mistakenly copies $(a-b)^2$ as a^2-b^2 .
nd $b = -3$, the amount of error is

- a) 25
- b) 0
- c) 52
- d) 42

n $3+6+9+12+15+ \dots + 45$ is

- a) 603
- b) 360
- c) 630
- d) 345

e and a square have same perimeter. Then

- a) Their areas equal
- b) The circle has greater area
- c) The square has greater area
- d) Circle's area = π times area of square

gonal of a square is $\sqrt{2}$ cm. Its area is

- a) 1 cm.
- b) 1 cm^2
- c) $2 \sqrt{2} \text{ cm}^2$
- d) 2 cm^2

ny prime numbers are there between 30 and 40 ?

- a) 0
- b) 1
- c) 2
- d) 3

ne is **not** correct?

- a) All rectangles are parallelograms.
- b) All squares are rhombuses
- c) All rectangles are not squares
- d) All rhombuses are squares.

ne of the following is equal to $(p-q-r)^2$?

- a) $(q-r-p)^2$
- b) $(r-p-q)^2$
- c) $(r+p-q)^2$
- d) $(q+r-p)^2$

of a substance is $\frac{1}{2}$ kg what is

the substance?

- a) 200g
- b) 200000g
- c) 20000g
- d) none of these

ve discounts of 20%, 20% and 10% is
a single discount of

- a) 44.2 %
- b) 42.4 %
- c) 24.2 %
- d) 24.4 %

on has two angles of 130° each and the

- a) 6
- b) 7

a triangle is 31.5 cm^2 and the base
er than the height. The length
s

ram with one pair of adjacent
al length is always

of an interior angle of a
ided polygon is

Pythagorean triplet since $3^2 + 4^2 = 5^2$.

f the following is not a Pythagorean triplet?

of the parallel sides of an isosceles
re 10 cm and 4 cm. If the distance between
sides is 4 cm, the perimeter
zium is (in cm)

the biggest square
a circle of radius 4 m is

irrational numbers is

ive when written in base 3?

led paper cone has a slant height 7 cm
radius 3.5 cm. If the cone is cut
line from a point P on the base edge to
and the paper is flattened out,
e the area of paper in sq.cm. ? ($\pi = 22/7$)

- a) 7 cm.
- b) 10 cm.
- c) 9 cm.
- d) 11 cm.

- a) a rhombus
- b) a square
- c) a trapezium
- d) a rectangle

- a) 144°
- b) 140°
- c) 154°
- d) 150°

- a) 7,24,25
- b) 8,15,17
- c) 13,35,37
- d) 11,60,61

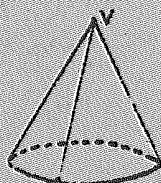
- a) 18
- b) 20
- c) 24
- d) 28

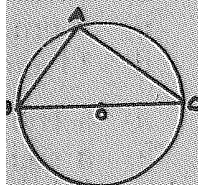
- a) 32 m^2
- b) 64 m^2
- c) 80 m^2
- d) None of these

- a) always irrational
- b) always rational
- c) never irrational
- d) sometimes rational

- a) 16
- b) 20
- c) 100
- d) 1000

- a) $77 \frac{1}{3}$
- b) $51 \frac{1}{3}$
- c) 38.5
- d) $25 \frac{2}{3}$

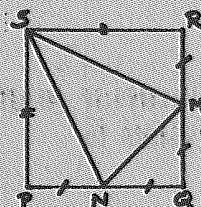




In the figure, $\triangle ABC$ is circumscribed such that the centre O lies on CB . If $m\angle ACB = 41$, then what is $m\angle BAO$?

- a) 82
- b) 41
- c) 139
- d) 49

se, ratio of square
to $\triangle MNS$ is



- a) 2 : 1
- b) 3 : 1
- c) 7 : 2
- d) 5 : 1

t have you to multiply $\frac{s}{t}$ to get $\frac{t}{s}$?

- a) $\frac{s^2}{t^2}$
- b) $\frac{t^2}{s^2}$
- c) $\frac{t-s}{s-t}$
- d) 1

$A = \{ 1, 2, 3 \}$
 $B = \{ 3, 2, 1 \}$
 $C = \{ 1, 2, 2, 3, 3, 3 \}$

- a) $A = B \neq C$
- b) $A \neq B = C$
- c) $A \neq B \neq C$
- d) $A = B = C$

ich of the statements is correct?

$z^2 = x^2 + y^2$, then

- a) $x = y$
- b) $x = -y$
- c) $x = 1, y = -1$
- d) at least one of the variables is zero.

s correct?

$\overline{2} = 2$, then $(x+2)^2$ equals

- a) $\sqrt{2}$
- b) 2
- c) 4
- d) 16

io between the areas of a square and a
on its diagonal is

- a) 1 : 2
- b) 1 : 2 $\sqrt{2}$
- c) 1 : 4
- d) 1 : 2

+ c = 0 and

- a) 3
- b) 2
- c) 1
- d) 0.5

+ c² = 1,

e value of $a^4 + b^4 + c^4$ is